

Polynomials in $X_1, X_2, \dots, X_n$ with coefficients in A , can be identified with formal power series, as follows: if $f \in A[X_1, X_2, \dots, X_n]$ and $f = f_0 + f_1 + \dots + f_m$, where each f_i is a form which is either zero or of degree i , then we identify f with the power series $(f_0, f_1, \dots, f_m, 0, 0, \dots)$. By this identification the polynomial ring $R = A[X_1, X_2, \dots, X_n]$ becomes a subring of the power series ring $S = A[[X_1, X_2, \dots, X_n]]$.

REMARK. If the ring A is the field of real or complex numbers, then the power series f which are *convergent* in a suitable neighborhood of the origin $X_1 = X_2 = \dots = X_n = 0$ become an object of study. It can be shown that the convergent power series form a subring S' of S (this subring obviously contains all the polynomials). Most of the results proved in this section (in particular, the Weierstrass preparation theorem and its consequences) hold also for S' .

Let $f = (f_0, f_1, \dots, f_q, \dots)$ be a *non-zero* power series. The smallest index q for which f_q is different from zero will be called the *order* of f and will be denoted by $\mathfrak{o}(f)$. If $i = \mathfrak{o}(f)$, then the form f_i is called the *initial form* of f . We agree to attach the order $+\infty$ to the element 0 of S .

THEOREM 1. If f and g are power series in $A[[X_1, X_2, \dots, X_n]]$, then

$$(3) \quad \mathfrak{o}(f+g) \geq \min\{\mathfrak{o}(f), \mathfrak{o}(g)\},$$

$$(4) \quad \mathfrak{o}(fg) \geq \mathfrak{o}(f) + \mathfrak{o}(g).$$

Furthermore, if A is an integral domain then also S is an integral domain and we have

$$(4') \quad \mathfrak{o}(fg) = \mathfrak{o}(f) + \mathfrak{o}(g).$$

PROOF. The proofs of (3) and (4) are straightforward and are similar to the proofs given for polynomial rings in Vol. I, Ch. I (see, for instance, I, § 18, proof of Theorem 11; the only difference in the proof is that now we have to use the initial forms rather than the homogeneous components of highest degree). As to (4'), we observe that if $f \neq 0$ and $g \neq 0$ then the product $f_i g_j$ of the initial forms of f and g is different from zero (since the polynomial ring $A[X_1, X_2, \dots, X_n]$ is an integral domain if A is an integral domain) and is the initial form of fg .

The power series of *positive* order form an ideal in S . This ideal is generated by $X_1, X_2, \dots, X_n$ and shall be denoted by $\mathfrak{x}$. For any integer $q \geq 1$, the ideal $\mathfrak{x}^q$ consists of those power series which have order $\geq q$. It follows that $\bigcap_{q=1}^{\infty} \mathfrak{x}^q = (0)$.

VII. POLYNOMIAL AND POWER SERIES RINGS

Among commutative rings, the polynomial rings in a finite number of indeterminates enjoy important special properties and are frequently used in applications. As they are also of paramount importance in Algebraic Geometry, polynomial rings have been intensively studied. On the other hand, rings of formal power series have been extensively used in "algebraic geometry" and have many properties which are parallel to those of polynomial rings. In the first section of this chapter we shall define formal power series and we shall show that the main properties of polynomial rings which have been derived in previous chapters (see, in particular, Vol. I, Ch. I, §§ 16–18) hold also for formal power series rings. In the later sections of this chapter we shall give deeper properties of polynomial rings and, whenever possible, the parallel properties of power series rings.

§ 1. **Formal power series.** Let A be a (commutative) ring with element 1 and let $R = A[X_1, X_2, \dots, X_n]$ be the polynomial ring in n indeterminates over A . By a *formal power series* in n indeterminates over A we mean an infinite sequence $f = (f_0, f_1, \dots, f_q, \dots)$ of homogeneous polynomials f_q in R , each polynomial f_q being either 0 or of degree q . We define addition and multiplication of two power series $f = (f_0, f_1, \dots, f_q, \dots)$ and $g = (g_0, g_1, \dots, g_q, \dots)$ as follows:

$$(1) \quad f+g = (f_0+g_0, f_1+g_1, \dots, f_q+g_q, \dots),$$

$$(2) \quad fg = (h_0, h_1, \dots, h_q, \dots), \text{ where } h_q = \sum_{i+j=q} f_i g_j.$$

It is easily seen that with these definitions of addition and multiplication the set S of all formal power series in n indeterminates over A becomes a *commutative ring*. This ring S , called the *ring of formal power series* in n indeterminates over A , shall be denoted by $A[[X_1, X_2, \dots, X_n]]$. The zero of S is the sequence $(0, 0, \dots)$, and $(1, 0, 0, \dots)$ is the multiplicative identity of S .

THEOREM 2. If $f = (f_0, f_1, \dots, f_q, \dots)$ is a power series, then f is a unit in S if and only if the element f_0 of A is a unit in A .

PROOF. If $fg = 1$, with $g = (g_0, g_1, \dots, g_q, \dots)$, then $f_0g_0 = 1$, and hence f_0 is a unit in A . Conversely, if f_0 is a unit in A , then we can find successively forms $g_0, g_1, \dots, g_q, \dots$, where g_q is either zero or a form of degree q , such that $g_0f_0 = 1, g_1f_0 + g_0f_1 = 0, \dots, g_qf_0 + g_{q-1}f_1 + \dots + g_0f_q = 0$. In fact, we have $g_0 = f_0^{-1}$. Assuming that $g_0, g_1, \dots, g_{q-1}$ have already been determined and that each g_i is either zero or a form of degree i ($0 \leq i \leq q-1$), we set $g_q = -f_0^{-1}(g_{q-1}f_1 + \dots + g_0f_q)$, and it is clear that g_q is then either zero or a form of degree q . If we now set $g = (g_0, g_1, \dots, g_q, \dots)$ then we find, by (2), that $fg = 1$. This completes the proof.

COROLLARY 1. If k is a field, then the units of the power series ring $k[[X_1, X_2, \dots, X_n]]$ are the power series of order 0. The ring $k[[X_1, X_2, \dots, X_n]]$ is a local ring, and the ideal $\mathfrak{z}$ generated by $X_1, X_2, \dots, X_n$ is its maximal ideal.

Everything follows directly from Theorem 2 except the assertion (implicit in the statement that $k[[X_1, X_2, \dots, X_n]]$ is a local ring) that $k[[X_1, X_2, \dots, X_n]]$ is noetherian. This will be proved later on in this section (see Theorem 4).

COROLLARY 2. If k is a field and $S = k[[X]]$ is the power series ring in one indeterminate, then $\mathfrak{z}$ is the principal ideal SX , and every ideal in S is a power of $\mathfrak{z}$. In other words, S is a discrete valuation ring, of rank 1, and its non-trivial ideals are the ideals SX^e .

Everything follows directly from Theorem 2 and from properties of p -adic valuations in unique factorization domains (p —an irreducible element; see VI, §9. Examples of valuations, 2), by observing that if f is a non-zero element of $k[[X]]$, of order q , then $f = X^qg$, where g is a unit.

The valuation of which $k[[X]]$ is the valuation ring is the one in which the value of any non-zero element f of $k[[X]]$ is the order $\mathfrak{o}(f)$ of f . Now, Theorem 1 shows that, more generally, if A is an integral domain and $S = A[[X_1, X_2, \dots, X_n]]$ is the power series ring in any number of indeterminates over A , then the mapping $f \rightarrow \mathfrak{o}(f)$ can be extended uniquely to a valuation of the quotient field of S (in general, however, S will not be the valuation ring of that valuation). If we denote by $\mathfrak{o}$ that valuation, then it is clear that the center of $\mathfrak{o}$ in S (see Ch. VI, §5) is the maximal ideal $\mathfrak{z}$ of S . We shall refer to this valuation $\mathfrak{o}$ as the $\mathfrak{z}$ -adic valuation of S (or of the quotient field of S). It is clear that the valuation $\mathfrak{o}$ is trivial on A , and hence we may assume that the residue field of $\mathfrak{o}$ contains the quotient field of A .

THEOREM 3. The quotients X_i/X_n , $i = 1, 2, \dots, n-1$, belong to the valuation ring of the $\mathfrak{z}$ -adic valuation $\mathfrak{o}$. If t_i denotes the residue of X_i/X_n in the valuation $\mathfrak{o}$, then $t_1, t_2, \dots, t_{n-1}$ are algebraically independent over A , and the residue field of $\mathfrak{o}$ is $k(t_1, t_2, \dots, t_{n-1})$, where k is the quotient field of A (in an integral domain).

PROOF. Since $\mathfrak{o}(X_i) = 1$, $i = 1, 2, \dots, n$, $\mathfrak{o}(X_i/X_n) = 0$, and the first assertion is proved. Let now $F(X_1, X_2, \dots, X_{n-1})$ be any non-zero polynomial in $n-1$ indeterminates, with coefficients in A , and let m be the degree of F . We set $g = g_m(X_1, X_2, \dots, X_n) = X_n^m F(X_1/X_n, X_2/X_n, \dots, X_{n-1}/X_n)$. Then g is a form of degree m in $X_1, X_2, \dots, X_n$, with coefficients in A . We have $\mathfrak{o}(g) = m = \mathfrak{o}(X_n^m)$, hence the $\mathfrak{o}$ -residue of the quotient g/X_n^m is different from zero. Since $g/X_n^m = F(X_1/X_n, X_2/X_n, \dots, X_{n-1}/X_n)$ and since $\mathfrak{o}$ is trivial on A , it follows that $F(t_1, t_2, \dots, t_{n-1}) \neq 0$, showing that $t_1, t_2, \dots, t_{n-1}$ are algebraically independent over A .

The field $k(t_1, t_2, \dots, t_{n-1})$ is contained in the residue field of $\mathfrak{o}$, and it remains to show that these two fields coincide. Let ξ be any element of the residue field of $\mathfrak{o}$, $\xi \neq 0$, and let f and g be elements of $A[[X_1, X_2, \dots, X_n]]$ such that ξ is the $\mathfrak{o}$ -residue of f/g . Since $\xi \neq 0$, we must have $\mathfrak{o}(f) = \mathfrak{o}(g)$. Let $\mathfrak{o}(f) = q$. Then both f/X_n^q and g/X_n^q have non-zero $\mathfrak{o}$ -residues, and the quotient of these two residues is ξ . It is therefore sufficient to show that the residues of f/X_n^q and g/X_n^q both belong to $k(t_1, t_2, \dots, t_{n-1})$. Consider, for instance, f/X_n^q . Let f_q be the initial form of f . Then $\mathfrak{o}(f - f_q) > q$, whence the $\mathfrak{o}$ -residue of f/X_n^q coincides with the $\mathfrak{o}$ -residue of f_q/X_n^q . Since $f_q(X_1, X_2, \dots, X_n)/X_n^q = f_q(X_1/X_n, X_2/X_n, \dots, X_{n-1}/X_n)$, the $\mathfrak{o}$ -residue of f_q/X_n^q is $f_q(t_1, t_2, \dots, t_{n-1})$ and belongs therefore to $A(t_1, t_2, \dots, t_{n-1})$. This completes the proof.

We note that the restriction of $\mathfrak{o}$ to the polynomial ring $R = A[X_1, X_2, \dots, X_n]$ is a prime divisor of the field $k(X_1, X_2, \dots, X_n)$, with the same residue field as $\mathfrak{o}$, and that if $n > 1$ then this prime divisor is of the second kind with respect to the ring R , its center in R being the point $X_1 = X_2 = \dots = X_n = 0$ (see Ch. VI, §14).

We now go back to the general case, in which A is an arbitrary ring. If we take the set of ideals $\mathfrak{z}^i$, $i = 0, 1, 2, \dots$, as a fundamental system of neighborhoods of the element 0 of S , then, by Theorem 1, S becomes a topological ring (S. L. Pontrjagin, *Topological groups*, p. 172). [Elements "near" a given element f_0 of S are those elements f for which $f - f_0$ has high order. Since we have $\mathfrak{o}((f - f_0) - (f_0 + g_0)) \geq \min\{\mathfrak{o}(f - f_0), \mathfrak{o}(g - g_0)\}$, $\mathfrak{o}(f - f_0) = \mathfrak{o}(f - g_0) + \mathfrak{o}(g - f_0) \geq \min\{\mathfrak{o}(f) + \mathfrak{o}(g - g_0), \mathfrak{o}(g_0) + \mathfrak{o}(f - f_0)\}$, and both $\mathfrak{o}(f)$ and $\mathfrak{o}(g_0)$ are non-negative

integers, it follows that $f+g$ and fg are near f_0+g_0 and f_0g_0 respectively provided f and g are sufficiently near f_0 and g_0 ; in other words, the ring operations in S are indeed continuous.] Note that in view of the relation $\bigcap_{q=1}^{\infty} \mathfrak{A}^q = (0)$, S is a Hausdorff space. As a matter of fact, the topology of S can be induced by a suitable metric in S ; namely, fix a real number $r > 1$ and define the distance $d(f, g)$ between any two elements f, g of S by the formula $d(f, g) = r^{-i}$, where $q = \mathfrak{o}(f - g)$.

The space S is complete, i.e., every Cauchy sequence $\{f^i\}$ of elements f^i of S converges in S . For let $f^i = (f_0^i, f_1^i, \dots, f_q^i, \dots)$. Since we are dealing with a Cauchy sequence, we must have $f^i = f^j$ for all $i, j \geq n(q)$, where $n(q)$ is an integer depending on q . We set $f_q = f_q^i$ for $i = n(q)$ and $f = (f_0, f_1, \dots, f_q, \dots)$. Then $\mathfrak{o}(f - f^i) > q$ if $i \geq \max\{n(0), n(1), \dots, n(q)\}$, showing that the sequence $\{f^i\}$ converges to f .

It follows in the usual way that if $\{f^i\}$ and $\{g^i\}$ are two Cauchy sequences, then

$$(5) \quad \lim_{i \rightarrow +\infty} (f^i + g^i) = \lim_{i \rightarrow +\infty} f^i + \lim_{i \rightarrow +\infty} g^i,$$

$$(5') \quad \lim_{i \rightarrow +\infty} f^i g^i = \lim_{i \rightarrow +\infty} f^i \cdot \lim_{i \rightarrow +\infty} g^i.$$

Let now $\{h^i\}$ be an infinite sequence of power series satisfying the sole condition that $\mathfrak{o}(h^i)$ tends to ∞ with i ; in other words, $\{h^i\}$ is a Cauchy sequence whose limit is the element 0 of S . Then the partial sums $f^i = h^0 + h^1 + \dots + h^i$ clearly form a Cauchy sequence. We express this by saying that the infinite series $h^0 + h^1 + \dots + h^i + \dots$ is convergent and we define the infinite sum $\sum_{i=0}^{\infty} h^i$ to be the limit f of the sequence $\{f^i\}$:

$$\sum_{i=0}^{\infty} h^i = \lim_{i \rightarrow +\infty} (h^0 + h^1 + \dots + h^i), \quad \text{if } \mathfrak{o}(h^i) \rightarrow +\infty.$$

It follows easily from the definition of $\sum_{i=0}^{\infty} h^i$ that this infinite sum is independent of the order in which the elements of the sequence $\{h^i\}$ are written. We have the usual rules of addition and multiplication of infinite series:

$$(6) \quad \sum_{i=0}^{\infty} g^i + \sum_{i=0}^{\infty} h^i = \sum_{i=0}^{\infty} (g^i + h^i),$$

$$(6') \quad \sum_{i=0}^{\infty} g^i \cdot \sum_{i=0}^{\infty} h^i = \sum_{i=0}^{\infty} (g^0 h^i + g^1 h^{i-1} + \dots + g^i h^0).$$

Relation (6) follows directly from (5). As to (6'), the left-hand side is, by (5'), the limit of the Cauchy sequence $\{g^0 h^i + g^1 h^{i-1} + \dots + g^i h^0\}$, where $g^0 h^i = \sum_{j=0}^{\infty} g^j \cdot \sum_{j=0}^{\infty} h^j$.

$= \sum_{i,j=0}^{\infty} g^i h^j$, while the right-hand side is the limit of the sequence $\{g^i h^i\}$, where $g^i h^i = \sum_{i,j=0}^{\infty} g^i h^j$. Hence $g^0 h^i - g^i h^0$ is a sum of terms $g^j h^j$ in which at least one of the integers i, j is $\geq |q/2|$. Since $\mathfrak{o}(g^i)$ and $\mathfrak{o}(h^i)$ tend to $+\infty$ with i , it follows that the two sequences $\{g^i h^i\}$ and $\{g^i h^0\}$ have the same limit, and this proves (6').

We note that (6') implies the distributive law

$$(6'') \quad h \sum_{i=0}^{\infty} g^i = \sum_{i=0}^{\infty} h g^i.$$

We also note that if we have $h^i = 0$ for all sufficiently large values of i , say for $i > m$, so that the sequence $\{h^i\}$ is essentially a finite sequence, then the infinite sum $\sum_{i=0}^{\infty} h^i$ coincides with the sum of the elements $h^0, h^1, \dots, h^m$ in the ring S .

We note that the inequality (3) generalizes to infinite sums, i.e., we have for any convergent series $\sum_{i=0}^{\infty} h^i$:

$$(7) \quad \mathfrak{o}\left(\sum_{i=0}^{\infty} h^i\right) \geq \min_i \{\mathfrak{o}(h^i)\}.$$

The notion of infinite sums allows us to write every power series $f = (f_0, f_1, f_2, \dots, f_q, \dots)$, where f_q is a form of degree q (or is zero), as an infinite sum; namely, we have

$$(8) \quad f = \sum_{i=0}^{\infty} f_i, \quad \text{or} \quad f = f_0 + f_1 + \dots + f_q + \dots.$$

In this form, f appears as an actual power series in $X_1, X_2, \dots, X_n$. The partial sums f^i are now polynomials $f_0 + f_1 + \dots + f_i$. Each monomial which occurs in any of the forms f_q will be called a *term* of the power series f .

In (8), every element f of S is represented as a limit of polynomials. Hence S is the closure of the polynomial ring $R = A[X_1, X_2, \dots, X_n]$, or—equivalently— R is everywhere dense in S . The following characterization of subrings of S which are everywhere dense in S will be used in the sequel:

LEMMA 1. A subring L of S is everywhere dense in S if and only if L has the following property: if f_q is any form in $X_1, X_2, \dots, X_n$ with coefficients in A , then L contains at least one element whose initial form is f_q .

PROOF. Assume that L is everywhere dense in S and let f_q be a form, of degree q . If n is an integer $> q$, L must contain an element f such that $\mathfrak{o}(f - f_q) \geq n$ (since f_q must be the limit of a sequence of elements

of L). Since $n > q$, the inequality $\mathbf{o}(f - f_n) \geq n$ implies that f_n is the initial form of f . Note that in this part of the proof we have not used the assumption that L is a subring of S .

Conversely, assume that L has the property stated in the lemma. Let f be any element of S . We shall construct an infinite sequence $\{f^i\}$, $f^i \in L$, such that $\mathbf{o}(f - f^i) \geq i$, whence $f = \lim f^i$. For $i = 0$ we simply set $f^0 = 0$. Let us assume that we have already defined the n elements $f^0, f^1, \dots, f^{n-1}$ in L and that we have then $\mathbf{o}(f - f^i) \geq i$ for $i = 0, 1, \dots, n-1$. If $\mathbf{o}(f - f^{n-1}) \geq n$, we set $f^n = f^{n-1}$. If $\mathbf{o}(f - f^{n-1}) = n-1$, let g_n be the initial form of $f - f^{n-1}$ and let h^{n-1} be some element of L whose initial form is g_n . If we set $f^n = f^{n-1} + h^{n-1}$, then $f^n \in L$, since L is a subring of S , and we have $\mathbf{o}(f - f^n) = \mathbf{o}(f - f^{n-1} - h^{n-1}) \geq n$, since both $f - f^{n-1}$ and h^{n-1} are of order $n-1$ and have the same initial form g_n . This completes the proof of the lemma.

We have seen in Vol. I, Ch. I that in any polynomial in $A[X_1, X_2, \dots, X_n]$ one can substitute for the indeterminates elements of any overring of A (see Vol. I, Ch. I, § 16, end of section). This operation of substitution cannot be performed for power series without further ado since infinite sums of power series have a meaning only if their partial sums form a Cauchy sequence (hence converge, in the formal sense explained above). Consider the power series ring $A[[Y_1, Y_2, \dots, Y_m]]$ in m indeterminates and m power series $f^1(X_1, X_2, \dots, X_n)$, $f^2(X_1, X_2, \dots, X_n)$, $\dots$, $f^m(X_1, X_2, \dots, X_n)$ in n indeterminates, over A . We assume that each of the m power series f^i is of order ≥ 1 .

Under this assumption we proceed to define $g(f^1, f^2, \dots, f^m)$, $g(Y_1, Y_2, \dots, Y_m)$ being any power series in $A[[Y_1, Y_2, \dots, Y_m]]$. Let $g = g_0 + g_1 + \dots + g_q + \dots$, g_i being either zero or a form of degree q in $Y_1, Y_2, \dots, Y_m$, with coefficients in A . Then $g(f^1, f^2, \dots, f^m)$ is defined as an element g^g of $A[[X_1, X_2, \dots, X_n]]$. Furthermore, by Theorem 1, g^g is a power series of order $\geq q$, since g_q is a form of degree q and since $\mathbf{o}(f^i) \geq 1$, $1 \leq i \leq m$. Hence the series $\sum_{q=0}^{\infty} g^g$ is defined as an element of $A[[X_1, X_2, \dots, X_n]]$. This power series $\sum_{q=0}^{\infty} g^g$ in $A[[X_1, X_2, \dots, X_n]]$ we call the result of substitution of $f^1, f^2, \dots, f^m$ into $g(Y_1, Y_2, \dots, Y_m)$, or the transform of $g(Y_1, Y_2, \dots, Y_m)$ by the substitution $Y_i \rightarrow f^i$. In symbols:

$$(9) \quad g(f^1, f^2, \dots, f^m) = \sum_{q=0}^{\infty} g_q(f^1, f^2, \dots, f^m) = \sum_{q=0}^{\infty} g^g.$$

For fixed $f^1, f^2, \dots, f^m$, (9) defines a mapping

$$(10) \quad g \rightarrow g(f^1, f^2, \dots, f^m), \quad g \in A[[Y_1, Y_2, \dots, Y_m]].$$

of $A[[Y_1, Y_2, \dots, Y_m]]$ into $A[[X_1, X_2, \dots, X_n]]$. We shall refer to (10) as the substitution mapping (relative to the substitution $Y_i \rightarrow f^i$). It follows easily from the rules (6) and (6') of addition and multiplication of infinite sums, that the substitution mapping (10) is a homomorphism. Furthermore, the mapping (10) is continuous (with respect to the topology introduced earlier in power series rings). To see this it is sufficient to show that if $\mathfrak{y}$ denotes the ideal generated in $A[[Y_1, Y_2, \dots, Y_m]]$ by $Y_1, Y_2, \dots, Y_m$, then the transform of $\mathfrak{y}^i$ by (10) is contained in $\mathfrak{x}^{i/\mathbf{o}}$, where $\mathbf{o}(i)$ tends to ∞ with i . This, however, is obvious, since from the definition of the substitution mapping it follows that if $g \in \mathfrak{y}^i$ then $g(f^1, f^2, \dots, f^m)$ belongs to $\mathfrak{x}^i$.

The image of the ring $A[[Y_1, Y_2, \dots, Y_m]]$ under the substitution mapping (10) is a subring of $A[[X_1, X_2, \dots, X_n]]$. We shall denote this subring by $A[[f^1, f^2, \dots, f^m]]$.

It is not difficult to see that any continuous homomorphism τ of $A[[Y_1, Y_2, \dots, Y_m]]$ into $A[[X_1, X_2, \dots, X_n]]$ is a substitution mapping. For let $\tau(Y_i) = f^i$. The continuity of τ requires that high powers of f^i belong to high powers of the ideal $\mathfrak{x}$. Hence $f^i \in \mathfrak{x}$, $i = 1, 2, \dots, m$. Now, let $g = g_0 + g_1 + \dots + g_q + \dots$ be any power series in $Y_1, Y_2, \dots, Y_m$. Since τ is a homomorphism we have $\tau(g_q) = g_q(f^1, f^2, \dots, f^m)$ and $\tau(\sum_{q=0}^i g_q) = \sum_{q=0}^i g_q(f^1, f^2, \dots, f^m)$. Since $g = \lim_{i \rightarrow \infty} (\sum_{q=0}^i g_q)$ and since τ is continuous, we must have

$$\tau(g) = \lim_{i \rightarrow \infty} \tau\left(\sum_{q=0}^i g_q\right) = \lim_{i \rightarrow \infty} \sum_{q=0}^i g_q(f^1, f^2, \dots, f^m),$$

i.e., $\tau(g) = g(f^1, f^2, \dots, f^m)$, in view of (9). This shows that τ is the substitution mapping relative to the substitution $Y_i \rightarrow f^i$.

In the special case $m = n$, the two rings $A[[Y_1, Y_2, \dots, Y_n]]$ and $A[[X_1, X_2, \dots, X_n]]$ coincide and we have $Y_i = X_i$. In this case, our substitution mapping defines a continuous homomorphism of the power series ring $A[[X_1, X_2, \dots, X_n]]$ into itself. We now describe a case in which this homomorphism is an automorphism.

LEMMA 2. Let $f^1, f^2, \dots, f^n$ be n power series in $A[[X_1, X_2, \dots, X_n]]$ such that the initial form of f^i is X_i ($1 \leq i \leq n$). Then the substitution mapping $\varphi: g(X_1, X_2, \dots, X_n) \rightarrow g(f^1, f^2, \dots, f^n)$ is an automorphism of the power series ring $A[[X_1, X_2, \dots, X_n]]$.

PROOF. We first show that the kernel of φ is zero. Let g be a non-zero power series in $A[[X_1, X_2, \dots, X_n]]$ and let g_i be its initial form. From (9) we find at once that $g(f^1, f^2, \dots, f^n) - g_i \in \mathfrak{x}^{n+1}$,

Hence $g(f^1, f^2, \dots, f^n) \neq 0$, and thus g is not in the kernel of φ . Observe that we have shown here the following: g and $g(f^1, f^2, \dots, f^n)$ have the same initial form.

We next show that φ maps $A[[X_1, X_2, \dots, X_n]]$ onto itself, i.e., that $A[[f^1, f^2, \dots, f^n]] = A[[X_1, X_2, \dots, X_n]]$. If $g_A(X_1, X_2, \dots, X_n)$ is any form, with coefficients in A , then we have just seen that $g(X_1, X_2, \dots, X_n)$ is the initial form of the element $g(f^1, f^2, \dots, f^n)$ of the ring $A[[f^1, f^2, \dots, f^n]]$. It follows therefore from Lemma 1 that the ring $A[[f^1, f^2, \dots, f^n]]$ is everywhere dense in $A[[X_1, X_2, \dots, X_n]]$, and in order to prove the lemma we have only to show that $A[[f^1, f^2, \dots, f^n]]$ is a closed subset of $A[[X_1, X_2, \dots, X_n]]$. Assume then that we have an element h , such that $h = \lim_{i \rightarrow \infty} g'(f^1, f^2, \dots, f^n)$, where $g'(X_1, X_2, \dots, X_n)$ is in $A[[X_1, X_2, \dots, X_n]]$. The order of $g'(f^1, f^2, \dots, f^n) - g(f^1, f^2, \dots, f^n)$ is the same as the order of $g'(X_1, X_2, \dots, X_n) - g(X_1, X_2, \dots, X_n) - g'(X_1, X_2, \dots, X_n)$. Hence $\{g'(X_1, X_2, \dots, X_n)\}$ must be a Cauchy sequence as well as $\{g(f^1, f^2, \dots, f^n)\}$. Let $g = \lim_{i \rightarrow \infty} g'(X_1, X_2, \dots, X_n)$. Since φ is continuous, it follows that $h = \varphi(g) = g(f^1, f^2, \dots, f^n)$, whence $h \in A[[f^1, f^2, \dots, f^n]]$. Q.E.D.

COROLLARY 1. Let $f^1, f^2, \dots, f^m$ be m power series in $A[[X_1, X_2, \dots, X_n]]$, $m \leq n$, such that the initial form of f^i is X_i . Then the substitution $X_i \rightarrow f^i$ defines an isomorphism $\varphi: g \rightarrow g(f^1, f^2, \dots, f^m)$ of $A[[Y_1, Y_2, \dots, Y_m]]$ into $A[[X_1, X_2, \dots, X_n]]$.

For the first part of the proof of Lemma 2 is independent of the assumption $m = n$.

COROLLARY 2. Let A be an integral domain and let $f^1, f^2, \dots, f^m$ be m power series in $A[[X_1, X_2, \dots, X_n]]$, $m \leq n$, such that the initial forms of the f^i are linearly independent linear forms $f_1^1, f_1^2, \dots, f_1^m$. Then the substitution mapping $\varphi: g(Y_1, Y_2, \dots, Y_m) \rightarrow g(f^1, f^2, \dots, f^m)$ is an isomorphism of $A[[Y_1, Y_2, \dots, Y_m]]$ into $A[[X_1, X_2, \dots, X_n]]$. If, furthermore, $m = n$, $Y_i = X_i$, $i = 1, 2, \dots, n$, and the determinant of the coefficients of the linear forms $f_1^1, f_1^2, \dots, f_1^n$ is a unit in A (in particular, if A is a field and the above determinant is $\neq 0$), then φ is an automorphism of $A[[X_1, X_2, \dots, X_n]]$.

If $g(Y_1, Y_2, \dots, Y_m)$ is the initial form of a non-zero element $g(Y_1, Y_2, \dots, Y_m)$ of $A[[Y_1, Y_2, \dots, Y_m]]$, then we find, as in the case of the lemma, that ε is also the order of $\varphi(g)$, since $\varphi(g) - g(f^1, f^2, \dots, f^m) \in \mathfrak{x}^{\varepsilon+1}$ and since in the integral domain A the linear independence of the linear forms $f_1^1, f_1^2, \dots, f_1^m$ and the non-vanishing of the form g , imply that $g(f^1, f^2, \dots, f^m)$ is different from zero and

has an initial form of degree ε (to see this it is sufficient to pass to the quotient field of A).

If $m = n$ and if the determinant of the coefficients of the linear forms $f_1^1, f_1^2, \dots, f_1^n$ is a unit in A , then, for each integer q , the linear substitution $X_i \rightarrow f_i^1$ maps onto itself the set of forms of degree q in $X_1, X_2, \dots, X_n$, with coefficients in A . It follows that also in the present case the ring $A[[f_1^1, f_1^2, \dots, f_1^n]]$ has the property of containing power series with arbitrarily preassigned initial forms, with coefficients in A , and the rest of the proof of the lemma is now applicable without any change.

THEOREM 4. If A is a noetherian ring, then the power series ring $A[[X]]$ is also noetherian.

PROOF. We give here a proof parallel to the second proof of Hilbert's basis theorem, cf. Vol. I, Ch. IV, § 1, i.e., a proof using the finite basis condition. Let $\mathfrak{A}$ be an ideal in $A[[X]]$. For any integer $i \geq 0$ denote by $L_i(\mathfrak{A})$ the set of elements of A consisting of 0 and of the coefficients of X^i in all elements of $\mathfrak{A}$ which are of order i . Then $L_i(\mathfrak{A})$ is an ideal in A , and the ideals $L_i(\mathfrak{A})$ constitute an ascending sequence. Their union $L(\mathfrak{A})$ is the ideal in A consisting of 0 and of the coefficients of the initial terms[†] of all non-zero elements of $\mathfrak{A}$. Since A is noetherian, $L(\mathfrak{A})$ has a finite basis $\{a_0, \dots, a_k\}$. We fix in $\mathfrak{A}$ a power series $F_i(X)$ whose initial term has a_i as coefficient. Denote by d the greatest integer among the orders of the series $F_i(X)$.

Now, for every $j < d$, let $\{b_{j1}, \dots, b_{j\mu(j)}\}$ be a finite basis of the ideal $L_j(\mathfrak{A})$, and let $G_{j\mu}(X)$ be a power series in $\mathfrak{A}$ whose initial term is $b_{j\mu} X^j$ ($1 \leq j \leq \mu(j)$). We shall prove that the ideal $\mathfrak{A}$ is generated by the series $F_i(X)$, $G_{j\mu}(X)$ ($1 \leq i \leq q$; $0 \leq j < d$; $1 \leq \mu(j) \leq \mu(j)$). We prove this in two steps:

(a) Let $\mathfrak{A}'$ be the ideal $(G_{j\mu}(X))$ generated by the elements $G_{j\mu}(X)$. We have $\mathfrak{A}' \subset \mathfrak{A}$. Every element $P(X)$ of $\mathfrak{A}'$ which has the order $j < d$ is congruent mod $\mathfrak{A}'$ to an element of $\mathfrak{A}$ which has order $\geq j+1$. In fact, the coefficient c of the initial term cX^j of $P(X)$ may be written in the form $c = \sum_{k=1}^{\mu(j)} c_k b_{k\mu}$ ($c_k \in A$). Thus $P(X) - \sum_{k=1}^{\mu(j)} c_k G_{k\mu}(X)$ is of order $\geq j+1$. It follows by successive applications of this result that every element of order $j < d$ of $\mathfrak{A}$ is congruent mod $\mathfrak{A}'$ to an element of $\mathfrak{A}$ of order $\geq d$. It remains to prove that any element of order $\geq d$ of $\mathfrak{A}$ of order $\geq d$.

[†] Since we are dealing now with power series in one variable, an initial form, of degree i , consists of just one term cX^i , $c \in A$.

$\mathfrak{A}$ is in the ideal $(F_1(X), G_1(X))$. We will even prove that such an element is in the ideal $(F_1(X), \dots, F_g(X))$.

(b) Let $P(X)$ be an element of $\mathfrak{A}$ of order $s \geq d$, and let cX^s be its initial term. We may write $c = \sum_{i=1}^s c_i a_i$ ($c_i \in A$). Thus $P(X) -$

$\sum_{i=1}^s c_i X^{s-\alpha(F_i)} F_i(X)$ is an element of order $\geq s+1$ of $\mathfrak{A}$. By successive applications of this result we get q sequences $\{c_i^{(n)}\}$ ($i=1, 2, \dots, q$; $n=s, s+1, \dots$; $c_i^{(s)}=c_i$) of elements of A such that, for every n , the power series

$$P(X) - \sum_{i=1}^q c_i^{(n)} X^{s-\alpha(F_i)} F_i(X)$$

is of order $> n$. As the exponents $j - \alpha(F_j)$ tend to infinity with j , each of the infinite sums $\sum_{j=1}^q c_j^{(n)} X^{j-\alpha(F_j)}$ converges and represents an element $s(X)$ of $A[[X]]$. Since the order of the power series $P(X) - \sum_{i=1}^q s_i(X) F_i(X)$ is greater than n for every n , this power series is 0, and we have $P(X) = \sum_{i=1}^q s_i(X) F_i(X)$. Q.E.D.

COROLLARY. The power series ring $A[[X_1, \dots, X_n]]$ in n indeterminates over a noetherian ring A (in particular, over a field, or over the ring of integers) is noetherian.

This follows from Theorem 4 by induction on n , since $A[[X_1, \dots, X_n]]$ is isomorphic to $A[[X_1, \dots, X_{n-1}]][[X_n]]$.

REMARK. A simple direct proof of the fact that $A[[X_1, \dots, X_n]]$ is noetherian may be given if one uses the fact that the polynomial ring $A[X_1, \dots, X_n]$ is noetherian. But, since this proof applies as well to a more general situation, we postpone it until the chapter on Local Algebra (see VIII, § 3, Example 1, p. 260). On the other hand we shall give later on in this section a proof that $k[[X_1, \dots, X_n]]$ is noetherian (k a field) using the Weierstrass' preparation theorem.

THEOREM 5. (Weierstrass preparation theorem) Let k be a field and let $F(X_1, \dots, X_n)$ be a non-invertible power series (i.e., a non-unit in $k[[X_1, X_2, \dots, X_n]]$) with coefficients in k . Suppose that $F(X_1, \dots, X_n)$ contains terms of the form aX_n^h with non-zero coefficient a , and denote by s (≥ 1) the smallest of all the exponents h having this property. Then for every power series $G(X_1, \dots, X_n)$ there exists a power series $U(X_1, \dots,$

X_n) and s power series $R_i(X_1, \dots, X_{n-1})$ in $X_1, \dots, X_{n-1}$ ($0 \leq i \leq s-1$) such that

$$(11) \quad G(X_1, \dots, X_n) = U(X_1, \dots, X_n) F(X_1, \dots, X_n) + \sum_{i=0}^{s-1} R_i(X_1, \dots, X_{n-1}) X_n^i.$$

The power series U and R_i are uniquely determined by G and F .

PROOF. For every power series $P(X_1, \dots, X_n)$ denote by $\tau(P)$ the sum of all terms in P which do not have X_n as a factor, and by $h(P)$ the factor of X_n in $P - \tau(P)$. In other words we have

$$(12) \quad P = \tau(P) + X_n h(P),$$

where $\tau(P)$, $h(P) \in k[[X_1, X_2, \dots, X_n]]$ and where, furthermore, $\tau(P)$ is a polynomial in X_n of degree $\leq s-1$, with coefficients in $k[[X_1, X_2, \dots, X_{n-1}]]$. Note that if the power series ring $k[[X_1, X_2, \dots, X_n]]$ is thought of as a vector space over the field k , then both operations τ and h are linear transformations in that vector space. By the definition of the integer s , $h(F)$ is a unit in $k[[X_1, X_2, \dots, X_n]]$ (see Theorem 2), and $\tau(F)$, regarded as a polynomial in X_n , has all its coefficients in the maximal ideal of the ring $k[[X_1, X_2, \dots, X_{n-1}]]$. We shall denote this maximal ideal by $\mathfrak{m}$.

The problem of finding power series U and $R_0, R_1, \dots, R_{s-1}$ such that (11) holds is equivalent to the problem of finding a power series U such that the following relation holds:

$$(11a) \quad h(G) = h(UF).$$

For if (11) holds, then $h(G - UF) = 0$, whence (11a) holds by linearity of h . Conversely, assume that U is a power series satisfying (11a). Then $h(G - UF) = 0$, whence $G - UF = \tau(G - UF)$ (by (12)), i.e., $G - UF$ is a polynomial in X_n of degree $\leq s-1$, with coefficients in $k[[X_1, X_2, \dots, X_{n-1}]]$, and so (11) holds.

We have $UF = U\tau(F) + X_n U h(F)$, and hence (11a) can be re-written as follows:

$$(11b) \quad h(G) = h(U\tau(F)) + U h(F),$$

and our problem is equivalent to finding a power series U satisfying (11b). Since $h(F)$ is a unit in $k[[X_1, X_2, \dots, X_n]]$ we shall try to construct the power series

$$(13) \quad V = U h(F).$$

We set

$$M = -\tau(F)[h(F)]^{-1}.$$

Then, by (13), $U\tau(F) = -MV$, and (11b) is equivalent to

$$(11c) \quad h(G) = -h(MV) + V.$$

For every power series P , denote by $m(P)$ the power series $h(MP)$. Notice that m is again a linear operation on power series. Furthermore, if P , considered as a power series in X_n over $k[[X_1, \dots, X_{n-1}]]$, has all its coefficients in some power m^j of the maximal ideal m , then $m(P)$ has all its coefficients in m^{j+1} . For convenience we set $H = h(G)$. With these notations condition (11c) may be written as follows:

$$(11d) \quad V = H + m(V).$$

Since m is linear, condition (11d) implies that $V = H + m(H + m(V)) = H + m(H) + m^2(V)$, and, by successive applications:

$$(11e) \quad V = H + m(H) + m^2(H) + \dots + m^q(H) + m^{q+1}(V),$$

for any integer $q \geq 0$.

The property of the operation m which we have just pointed out above shows that $m^q(H)$ is at least of order j , and $m^{q+1}(V)$ is at least of order $q+1$. Thus the infinite sum $H + m(H) + m^2(H) + \dots + m^q(H) + \dots$ converges, and, if a power series V satisfying (11d) exists, it must therefore be the series

$$(15) \quad V = H + m(H) + m^2(H) + \dots + m^q(H) + \dots,$$

and this proves the uniqueness of V , whence of U and of the R_i .

We now prove that the series V given by (15) satisfies condition (11d). Let us write $V = H + m(H) + \dots + m^q(H) + W_q$. The coefficients of W_q (W_q being considered as a power series in X_n) are all in m^{q+1} . Then, since m is linear,

$$V - H - m(V) = H + \dots + m^q(H) + W_q - H - m(H) - \dots - m^{q+1}(H) - m(W_q) = W_q - m^{q+1}(H) - m(W_q).$$

Thus all coefficients of $V - H - m(V)$ are in m^{q+1} . As this is true for every q , we have $V - H - m(V) = 0$, and condition (11d) holds. This proves the existence of V , whence also of U and of the R_i .

REMARK. In the next chapter we shall give a somewhat shorter proof of the Weierstrass preparation theorem, based upon the properties of complete local rings. An advantage of the proof given here is that the questions of existence and unicity are treated simultaneously. A more

substantial advantage is that the method of majorants is easily applicable to the resolving formula (11e), with the result that if F and G are convergent power series over the field of real or complex numbers, then the series V , U and the R_i are also convergent. To show this we open now a brief digression on the preparation theorem for convergent power series.

In the case of convergent power series over the field k of real or complex numbers, the proof of the Weierstrass preparation theorem runs as follows. We recall that a power series

$$F(X_1, \dots, X_n) = \sum_{\alpha} a_{\alpha_1, \dots, \alpha_n} X_1^{\alpha_1} \dots X_n^{\alpha_n}$$

is said to be convergent if there exists a neighborhood N of the origin in k^n such that the series $\sum_{\alpha} a_{\alpha_1, \dots, \alpha_n} z_1^{\alpha_1} \dots z_n^{\alpha_n}$ is absolutely convergent for every $(z_1, \dots, z_n) \in N$. Then there exist positive real numbers μ and ρ such that $|a_{\alpha_1, \dots, \alpha_n}| \leq \mu \rho^{-\alpha_1 - \dots - \alpha_n}$. Conversely, the existence of two such real numbers implies that $\sum_{\alpha} a_{\alpha_1, \dots, \alpha_n} z_1^{\alpha_1} \dots z_n^{\alpha_n}$ converges in the neighborhood N of 0 defined by $|z_i| < \rho$ ($i=1, \dots, n$). It is easily seen that the convergent power series in $k[[X_1, \dots, X_n]]$ form a subring of $k[[X_1, \dots, X_n]]$, and that a convergent power series with a constant term $\neq 0$ admits as inverse a convergent power series. A series $\sum_{\alpha} a_{\alpha_1, \dots, \alpha_n} X_1^{\alpha_1} \dots X_n^{\alpha_n}$ with real positive coefficients is said to be a majorant of $\sum_{\alpha} b_{\alpha_1, \dots, \alpha_n} X_1^{\alpha_1} \dots X_n^{\alpha_n}$ if

$|b_{\alpha_1, \dots, \alpha_n}| \leq a_{\alpha_1, \dots, \alpha_n}$ for all $\alpha_1, \dots, \alpha_n$. It is clear that, in order to prove the convergence of a power series F , it is sufficient to prove that a majorant of F converges. The inequality $|a_{\alpha_1, \dots, \alpha_n}| \leq \mu \rho^{-\alpha_1 - \dots - \alpha_n}$ means that $\mu \left(1 - \frac{X_1}{\rho}\right) \dots \left(1 - \frac{X_n}{\rho}\right)$ is a majorant of $\sum_{\alpha} a_{\alpha_1, \dots, \alpha_n} X_1^{\alpha_1} \dots X_n^{\alpha_n}$.

In order to extend the Weierstrass preparation theorem to convergent power series, it is sufficient to prove that, if the series M and H are convergent, then

$$V = H + m(H) + \dots + m^q(H) + \dots,$$

is convergent (same notations as in the proof of Theorem 5). We notice that the coefficients of V are polynomials with positive integral coefficients in the coefficients of M and H . Thus, if we replace M and

† See Bochner-Martin, "Several complex variables," Princeton (1948), Chap. II.

H by majorants M' and H' , and assuming that M' is of positive order, then the power series

$$V' = H' + m'(H') + \dots + m^q(H') + \dots$$

(where the operation m' is defined by $m'(P) = h(M'/P)$) is a majorant of V . We may take

$$H'_1 = \frac{\mu}{\left(1 - \frac{X_1}{\rho}\right) \dots \left(1 - \frac{X_{n-1}}{\rho}\right)},$$

$$M' = \frac{\mu(X_1 + \dots + X_{n-1})}{\left(1 - \frac{X_1}{\rho}\right) \dots \left(1 - \frac{X_{n-1}}{\rho}\right)}.$$

(For the second one we write $M = N_1 X_1 + \dots + N_{n-1} X_{n-1}$ and we major separately each one of the series N_i .) Instead of H'_1 , we take as majorant of H the series

$$H' = \frac{\mu}{\left(1 - \frac{X_1}{\rho}\right) \dots \left(1 - \frac{X_{n-1}}{\rho}\right)} \varphi\left(\frac{X_n}{\rho}\right),$$

where $\varphi(X)$ is a series in one variable, majoring $\frac{1}{1-X}$ and enjoying properties which we are going to describe.

We notice that the operation m' is not only additive, but *linear over* $k[[X_1, \dots, X_{n-1}]]$. We thus have

$$m'(H') = \frac{\mu^2(X_1 + \dots + X_{n-1})}{\left(1 - \frac{X_1}{\rho}\right)^2 \dots \left(1 - \frac{X_{n-1}}{\rho}\right)^2} h\left(\varphi\left(\frac{X_n}{\rho}\right)\right) \left(1 - \frac{X_n}{\rho}\right).$$

We set $X = \frac{X_n}{\rho}$. The series $V' = H' + m'(H') + \dots + m^q(H') + \dots$ will

be very easy to compute if $h\left(\frac{\varphi(X)}{1-X}\right)$ is a scalar multiple of the series $\varphi(X)$. By definition of the operation h , this is true if there exist a polynomial $P_{s-1}(X)$ of degree $\leq s-1$ and a real number λ such that

$$\frac{\varphi(X)}{1-X} = P_{s-1}(X) + \lambda X^s \varphi(X).$$

Thus $\varphi(X)$ must be a rational function:

$$\varphi(X) = \frac{(1-X)P_{s-1}(X)}{1-\lambda X^s + \lambda X^{s+1}}.$$

We take $\lambda = 2^{s+1}$, and notice that the denominator $1 - 2^{s+1}X + 2^{s+1}X^{s+1}$ factors into $(1-2X)(1+2X+2^2X^2+\dots+2^{s-1}X^{s-1}-2^sX^s)$. The second factor takes the value 1 for $X=0$ and -1 for $X=1$. Therefore it admits a positive root $1/\alpha$ ($\alpha > 1$). Thus the denominator $1 - 2^{s+1}X + 2^{s+1}X^{s+1}$ may be written in the form $(1-2X)(1-\alpha X)P_{s-1}(X)$, where $P_{s-1}(X)$ is a polynomial of degree $s-1$. We choose $P_{s-1}(X)$ to be just this polynomial $P_{s-1}(X)$. We then have

$$\varphi(X) = \frac{1-X}{1-2X} \frac{1}{1-\alpha X},$$

and thus for this choice of $\varphi(X)$ we will have $h(\varphi(X))/(1-X) = 2^{s+1}\varphi(X)$. As it is a rational function, this power series $\varphi(X)$ is convergent. Since

$$\frac{1-X}{1-2X} = \frac{1}{2} + \frac{1}{2} \frac{1}{1-2X}$$

the power series expansion of $\varphi(X)$ is

$$\frac{1}{2}(2+2X+4X^2+\dots+2^sX^s+\dots)(1+\alpha X+\dots+\alpha^sX^s+\dots).$$

Except for the constant term (which is equal to 1), the coefficient of X^n is $\alpha^n + \alpha^{n-1} + 2\alpha^{n-2} + \dots + 2^{n-1}$; since it is obviously > 1 , $\varphi(X)$ is a majorant of $1/(1-X) = 1 + X + \dots + X^n + \dots$.

This being so, if we set $A = \mu \left(1 - \frac{X_1}{\rho}\right) \dots \left(1 - \frac{X_{n-1}}{\rho}\right)$ and $B = \mu(X_1 + \dots + X_{n-1}) \left(1 - \frac{X_1}{\rho}\right) \dots \left(1 - \frac{X_{n-1}}{\rho}\right)$ and if we notice that, for every power series $\psi(X)$ (where $X = X_n/\rho$), we have

$$m'(\psi(X)) = h\left(\frac{\psi(X)}{1-X}\right) = Bh\left(\frac{\psi(X)}{1-X}\right),$$

we get $m'(H') = m'(A\varphi(X)) = ABh(\varphi(X))/(1-X) = 2^{s+1}AB\varphi(X)$. Hence, by repeated applications,

$$m'^2(H') = m'(2^{s+1}AB\varphi(X)) = 2^{s(s+1)}AB^2\varphi(X)$$

and $m^q(H') = 2^{q(s+1)s}AB^q\varphi(X)$ for every q . Then the computation of the infinite sum $V' = H' + m'(H') + \dots + m^q(H') + \dots$, reduces to the computation of the sum of a geometric series:

$$V' = A\varphi(X) + A2^{s+1}B\varphi(X) + A2^{2(s+1)}B^2\varphi(X) + \dots$$

Hence

$$\begin{aligned} V' &= A_{\rho}(X) \frac{1}{1-2^{s+1}B} \\ &= \frac{\mu_{\rho}(X_1/\rho) \left(1 - \frac{X_1}{\rho}\right) \cdots \left(1 - \frac{X_{n-1}}{\rho}\right) \left(1 - \frac{X_n}{\rho}\right)}{\left(1 - 2^{s+1}\mu(X_1/\rho) + \cdots + X_{n-1}/\rho\right)} \\ &= \frac{\mu_{\rho}\left(1 - \frac{X_n}{\rho}\right)}{\left(1 - 2^{s+1}\mu\left(1 - \frac{X_1}{\rho}\right) \cdots \left(1 - \frac{X_{n-1}}{\rho}\right) - 2^{s+1}\mu(X_1 + \cdots + X_{n-1})\right)}. \end{aligned}$$

Since V' is a rational function, this is a convergent power series. This proves the preparation theorem in the case of convergent power series.

A power series $F(X_1, X_2, \dots, X_n)$ which contains a term cX_1^i which is a power of X_n with non-zero coefficient c , is said to be *regular* in X_n . To say that $F(X_1, X_2, \dots, X_n)$ is regular in X_n is equivalent to saying that $F(0, 0, \dots, 0, X_n)$ is different from zero.

COROLLARY 1. Let $F(X_1, X_2, \dots, X_n)$ be a power series in $S = k[[X_1, X_2, \dots, X_n]]$ which is regular in X_n (k a field) and let the order s of the power series $F(0, 0, \dots, 0, X_n)$ be ≥ 1 (in other words, it is assumed that F is not a unit).[†] Then there exist power series $E(X_1, X_2, \dots, X_n)$, $R_s(X_1, X_2, \dots, X_{n-1})$ ($i = 0, 1, \dots, s-1$) such that

$$\begin{aligned} (16) \quad & F(X_1, X_2, \dots, X_n) \\ &= E(X_1, X_2, \dots, X_n)[X_n^s + R_{s-1}(X_1, X_2, \dots, X_{n-1})X_n^{s-1} + \cdots \\ &\quad + R_0(X_1, X_2, \dots, X_{n-1})]. \end{aligned}$$

The power series E, R_i are uniquely determined by F ; E is a unit, and none of the R_i is a unit.

For if we apply Theorem 5 to the power series $G = -X_n$, we find

$$\begin{aligned} X_n^s + R_{s-1}(X_1, X_2, \dots, X_{n-1})X_n^{s-1} + \cdots + R_0(X_1, X_2, \dots, X_{n-1}) \\ = -U(X_1, X_2, \dots, X_n)F(X_1, X_2, \dots, X_n). \end{aligned}$$

Setting $X_1 = X_2 = \cdots = X_{n-1} = 0$ in this identity we obtain on the right-hand side a power series in X_n which has order $\geq s$. Hence $R_i(0, 0, \dots, 0) = 0$, $0 \leq i \leq s-1$, and no $R_i(X_1, X_2, \dots, X_{n-1})$ is a unit. It follows at the same time that $U(0, 0, \dots, 0, X_n)$ must be of order

[†] The corollary holds trivially also if F is a unit; in (16) we have then $F = E$, while the expression in the square brackets is the element 1.

zero, whence $U(X_1, X_2, \dots, X_n)$ is a unit. If we now set $E = U^{-1}$, we have (16). The unicity of E and of the R_i also follows from Theorem 5 in the special case $G = -X_n$.
The polynomial (in X_n)

$$(17) \quad F^* = X_n^s + R_{s-1}(X_1, X_2, \dots, X_{n-1})X_n^{s-1} + \cdots + R_0(X_1, X_2, \dots, X_{n-1})$$

in (16) is called the *distinguished pseudo-polynomial* associated with F ; it is defined only if F is regular in X_n , and its degree s (in X_n) is equal to the order of the power series $F(0, 0, \dots, 0, X_n)$. The relation (16) shows that F and F^* are associates in S .[†]

Note that F^* has the following two properties: (a) it is a monic polynomial in X_n ; (b) its coefficients, other than the leading coefficient, are power series in $X_1, X_2, \dots, X_{n-1}$ which belong to the maximal ideal of $k[[X_1, X_2, \dots, X_{n-1}]]$. Before deriving other consequences of Weierstrass' preparation theorem, we point out the following consequence of (a) and (b): if S^* denotes the ring

$$k[[X_1, X_2, \dots, X_{n-1}]][[X_n]],$$

then

$$(18) \quad SF^* \cap S^* = S^* F^*.$$

We have to show the following: if $H^* = hF^*$, with $H^* \in S^*$ and $h \in S$, then $h \in S^*$. Let $h = \sum_{q=0}^{\infty} h_q(X_1, X_2, \dots, X_{n-1})X_n^q$ and let $s+m$ be the degree of H^* in X_n . Expressing the fact that hF^* , regarded as a power series in X_n , is actually a polynomial of degree $s+m$, we find

$$(19) \quad h_q + h_{q+1}R_{s-1} + \cdots + h_{q+s}R_0 = 0, \quad q > m.$$

Since the R_i all belong to $\mathfrak{m}$, it follows from (19) that $h_q \in \mathfrak{m}$ if $q > m$. But then again (19) shows that $h_q \in \mathfrak{m}^s$ if $q > m$. By repeated application of this argument we find that $h_q \in \bigcap_{i=1}^{\infty} \mathfrak{m}^i$, whence $h_q = 0$ for all $q > m$. Thus h is a polynomial in X_n (of degree m), showing that $h \in S^*$.

Since F and F^* are associates in S we have $SF = SF^*$. Then (18) shows that the residue class ring S/SF contains S^*/S^*F^* as a subring.

COROLLARY 2. The rings S/SF and S^*/S^*F^* coincide.

For if G is any element of S then Theorem 5 shows that G is congruent mod F to an element of S^* .

[†] Note that the distinguished pseudo-polynomial of a unit F is the element 1.

The following lemma shows that every non-zero power series in $k[[X_1, X_2, \dots, X_n]]$ may be construed to be regular in X_n . More precisely, we have

LEMMA 3. *If $F(X_1, X_2, \dots, X_n)$ is a non-zero power series in $k[[X_1, X_2, \dots, X_n]]$ (k, a field), then there exists an automorphism φ of $k[[X_1, X_2, \dots, X_n]]$ such that $\varphi(F)$ is regular in X_n .*

PROOF. We assume first that k is an infinite field. Let f_i be the initial form of F . Since k is infinite we can find elements $a_1, a_2, \dots, a_{n-1}$ in k such that $f_i(a_1, a_2, \dots, a_{n-1}, 1) \neq 0$. Then we may use the linear substitution $X_i \rightarrow X_i + a_i X_n$ ($i=1, 2, \dots, n-1$). $X_n \rightarrow X_n$ (compare with the normalization lemma of Vol. I, Ch. V, § 4, Theorem 8). By Lemma 2, Corollary 2, the corresponding substitution mapping φ is an automorphism. Furthermore, the initial form of $\varphi(F)$ contains the term $f_i(a_1, a_2, \dots, a_{n-1}, 1)X_n^i$. Hence $\varphi(F)$ is regular in X_n .

We now give a proof which is also valid for finite fields and which will show the existence of exponents u_j ($j=1, \dots, n-1$) such that the automorphism φ defined by $\varphi(X_n) = X_n$, $\varphi(X_j) = X_j + X_n^{u_j}$ has the required property, i.e., is such that $F(X_n^{u_1}, \dots, X_n^{u_{n-1}}, X_n) \neq 0$. We order lexicographically the monomials which appear in F with non-zero coefficients. Let $X_1^{a_1} \dots X_n^{a_n}$ be the smallest one. Then, if $X_1^{b_1} \dots X_n^{b_n}$ is another monomial which actually appears in F , we have, either $b_1 > a_1$, or $b_1 = a_1$ and $b_2 > a_2$, or $b_1 = a_1, \dots, b_{n-1} = a_{n-1}$ and $b_n > a_n$. The corresponding monomials in $F(X_n^{u_1}, \dots, X_n^{u_{n-1}}, X_n)$ have $u_1 a_1 + u_2 a_2 + \dots + u_{n-1} a_{n-1} + a_n$ and $u_1 b_1 + u_2 b_2 + \dots + u_{n-1} b_{n-1} + b_n$ as exponents. If we take $u_{n-1} > a_n$, $u_{n-2} > u_{n-1} a_{n-1} + a_n, \dots, u_1 > u_2 a_2 + \dots + u_{n-1} a_{n-1} + a_n$, then we get $u_1 b_1 + \dots + u_{n-1} b_{n-1} + b_n > u_1 a_1 + \dots + u_{n-1} a_{n-1} + a_n$; in fact, if the index i is defined by the condition $a_i = b_i, \dots, a_{i-1} = b_{i-1}, a_i < b_i$, then the difference $u_1 b_1 + \dots + u_{i-1} b_{i-1} + u_i a_i + \dots + u_n a_n$ of the two above exponents is $u_i(b_i - a_i) + u_{i+1}(b_{i+1} - a_{i+1}) + \dots + b_n - a_n$. The first term is $\geq u_i$, whereas the remainder is $\geq -(u_{i+1} a_{i+1} + \dots + a_n)$, and thus the difference of the two exponents is > 0 since $u_i > u_{i+1} a_{i+1} + \dots + a_n$. In other words, in $F(X_n^{u_1}, \dots, X_n^{u_{n-1}}, X_n)$ the monomial with exponent $u_1 a_1 + \dots + u_n a_n$ cannot be cancelled by any other, and hence $F(X_n^{u_1}, \dots, X_n^{u_{n-1}}, X_n) \neq 0$.

COROLLARY. *Given any finite set of non-zero power series $F_1, F_2, \dots, F_h$ in $k[[X_1, X_2, \dots, X_n]]$, there exists an automorphism φ of $k[[X_1, X_2, \dots, X_n]]$ such that each of the h power series $\varphi(F_i)$ is regular in X_n .*

It is sufficient to apply the lemma to the product $F_1 F_2 \dots F_h$.

We give now a second proof of the fact that $k[[X_1, \dots, X_n]]$ is

noetherian. This proof can be applied *verbatim* to rings of convergent power series.

THEOREM 4'. *If k is a field, the formal power series ring $k[[X_1, \dots, X_n]]$ is noetherian.*

We prove by induction on n that every ideal $\mathfrak{A}$ in $k[[X_1, \dots, X_n]]$ has a finite basis (the cases $n=0$ and $n=1$ being trivial). We may suppose that $\mathfrak{A} \neq (0)$. By replacing, if necessary, $\mathfrak{A}$ by an automorphic image $\varphi(\mathfrak{A})$, we may suppose that $\mathfrak{A}$ contains a power series F which is regular in X_n (Lemma 3). For every G in $\mathfrak{A}$, we may write then $G = UF + \sum_{i=1}^{n-1} R_i X_i$ (Theorem 5). In other words, if we denote by S' the power series ring $k[[X_1, \dots, X_{n-1}]]$, we have $\mathfrak{A} = (F) + \mathfrak{A} \cap (S' + S'X_n + \dots + S'X_n^{n-1})$. As S' is a noetherian ring, by hypothesis, $\mathfrak{A} \cap (S' + S'X_n + \dots + S'X_n^{n-1})$ is a finitely generated S' -module, since it is a submodule of the finitely generated S' -module $S' + S'X_n + \dots + S'X_n^{n-1}$. A finite system of generators of $\mathfrak{A} \cap (S' + \dots + S'X_n^{n-1})$ will thus constitute, together with F , a finite basis of $\mathfrak{A}$. Q.E.D.

We end this section with another application of the Weierstrass preparation theorem. The proof we will give can be applied almost *verbatim* to rings of convergent power series.

THEOREM 6. *If k is a field, the formal power series ring $k[[X_1, \dots, X_n]]$ is a unique factorization domain.*

PROOF. We proceed by induction on n , the cases $n=0$ and $n=1$ being trivial. Since $k[[X_1, \dots, X_n]]$ is noetherian, we have to prove that, if F is an irreducible power series, then the principal ideal (F) is prime; in other words, we have to prove that, if $GH \in (F)$, then either G or H is a multiple of F . Let us write $GH = DF$. By replacing, if necessary, the series F, G, H, D by automorphic images $\alpha(F), \alpha(G), \alpha(H), \alpha(D)$, we may suppose that F, G, H, D are regular in X_n (corollary to Lemma 3). We denote by F', G', H', D' the distinguished pseudo-polynomials associated with F, G, H, D (Corollary 1 to Theorem 5). Since the power series $G'H'$ differs from GH by a unit only, and since it is a distinguished pseudo-polynomial of the right degree in X_n , it is the distinguished pseudo-polynomial associated with GH . Similarly $D'F'$ is the distinguished pseudo-polynomial associated with DF . As $DF = GH$, we have $D'F' = G'H'$, since the distinguished pseudo-polynomial associated with a given power series is unique.

Now, F' is an irreducible element of $k[[X_1, X_2, \dots, X_{n-1}]]$. In fact, assume that $g(X_1, X_2, \dots, X_{n-1}; X_n)$ is a factor of F' in $k[[X_1, X_2, \dots, X_{n-1}]]$, not a unit in this latter ring. The leading coefficient of g is a unit in $k[[X_1, X_2, \dots, X_{n-1}]]$ since the leading

coefficient of F' is 1 (both g and F' being regarded as polynomials in X_n). Therefore g must be of positive degree in X_n (for g is not a unit in $k[[X_1, X_2, \dots, X_{n-1}]]X_n$) and also $g(0, 0, \dots, 0; X_n)$ must be of positive degree in X_n . Consequently $g(0, 0, \dots, 0; X_n)$ is of the form cX_n^h , $h \geq 1$, $c \in k$, $c \neq 0$, since $F'(0, 0, \dots, 0; X_n)$ is also of this form. This shows that $g(0, 0, \dots, 0; 0) = 0$, i.e., that $g(X_1, X_2, \dots, X_{n-1}, X_n)$ is a non-unit in $k[[X_1, X_2, \dots, X_{n-1}]]X_n$. Since F is an irreducible element of $k[[X_1, X_2, \dots, X_{n-1}]]X_n$, we have proved that F' is also an irreducible element of $k[[X_1, X_2, \dots, X_{n-1}]]X_n$. By the induction hypothesis, $k[[X_1, X_2, \dots, X_{n-1}]]$ is a UFD, whence also $k[[X_1, X_2, \dots, X_{n-1}]]X_n$ is also a UFD. (Vol. I, Ch. I, § 18, Theorem 13.) Thus, from $D'F' = G'H'$ we deduce that either G' or H' is a multiple of F' in $k[[X_1, X_2, \dots, X_{n-1}]]X_n$. Hence, *a fortiori*, either G' or H' is a multiple of F' in $k[[X_1, X_2, \dots, X_{n-1}]]X_n$. Since F' , G' and H' differ from F , G and H only by unit factors in $k[[X_1, X_2, \dots, X_{n-1}]]X_n$, we conclude that either G or H is a multiple of F . This completes the proof.

COROLLARY. *If $F(X_1, X_2, \dots, X_n)$ is a power series which is regular in X_n and is an irreducible element of $k[[X_1, X_2, \dots, X_n]]$, then the quotient field of the residue class ring S/SF is a simple algebraic extension of the quotient field of $k[[X_1, X_2, \dots, X_{n-1}]]$.*

This follows immediately from Corollary 2 of Theorem 5.

§ 2. Graded rings and homogeneous ideals. Let A be a ring and let $R = A[X_1, X_2, \dots, X_n]$ be the polynomial ring over A , in n indeterminates. Every element F in R can be written in the form of a finite sum $F = F_0 + F_1 + \dots + F_j + \dots$, where F_j is either zero or a form of degree j . The form F_j is called the *homogeneous component of degree j* of F . The product of two homogeneous polynomials f and g is again homogeneous, and if $fg \neq 0$ then $\partial(fg) = \partial(f) + \partial(g)$ ($\partial = \text{degree}$). The homogeneous polynomials of a given degree q form, together with zero, an additive group and a finite A -module R_q . We have

$$(1) \quad R_q R_{q'} \subset R_{q+q'},$$

$$(2) \quad R = \sum_{q=-\infty}^{+\infty} R_q, \quad \text{the sum being direct,}$$

and R is an infinite (weak) direct sum (see Vol. I, Ch. III, § 12^{bis}) of the subgroups R_q :

where, in the present case of polynomial rings, we have $R_q = (0)$ if $q < 0$.

An ideal $\mathfrak{A}$ in R is said to be *homogeneous* if the relation $F \in \mathfrak{A}$ implies that all homogeneous components of F are in $\mathfrak{A}$.